

Kannada Agricultural Schemes QA System - RAG-based Question Answering System for Government Agricultural Schemes in Kannada

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Abstract—One of the most serious information challenges farmers experience in rural Karnataka is that the information on more than 80 government agricultural schemes is available only in Kannada, and is found in isolated government portals and other platforms. In this paper, we compare two question answering systems that use Retrieval-Augmented Generation (RAG) to enable access to scheme information in the farmer’s native language. 80 scheme documents were collected covering 10 mandatory topics per scheme and a golden dataset of 794 verified Kannada question-answer pairs was created. There are two experimental phases to consider, and we will explore both. In Phase 1, all three language models (Sarvam-2B, IndicBERT, XLM-RoBERTa) are evaluated on the entire golden set with a few transformations, with XLM-RoBERTa attaining the highest zero-shot BLEU score of 43.38 and BERTScore-F1 of 0.858. In Phase 2, a complete RAG pipeline with cross-encoder reranking and FAISS dense retrieval is tested on five models (MuRIL, XLM-RoBERTa, Sarvam-2B, IndicBARTSS, and Indic-GPT) with five metrics (BLEU, BERTScore-F1, Answer Similarity, Context Relevance, and Faithfulness). XLM-RoBERTa gets the highest RAG BLEU (4.89) and BERTScore-F1 (0.683). The lightweight seq2seq model IndicBARTSS (245M-parameter) outperforms its GPU counterparts in terms of Faithfulness (0.912).

Index Terms—Retrieval-Augmented Generation, Kannada NLP, Agricultural Question Answering, Large Language Models, FAISS, BERTScore, Low-Resource Languages, Indic Languages

I. INTRODUCTION

The agriculture sector is the backbone of the rural economy of Karnataka, where more than five million farming families are involved. Together, the central and state governments run over 80 schemes on a range of areas, including crop insurance, direct income support, input subsidies, credit facilitation, mechanisation support, and irrigation. However, farmers face a recurring challenge in obtaining reliable data on schemes due to the unavailability of such information in their local language, which is formal Kannada, and is located in isolated government portals. This lack of information symmetry has a direct impact on the uptake of schemes and loss of the benefits to which they are entitled, for a population with low levels of digital literacy.

A viable approach to bridging this gap is the use of natural language processing, specifically question answering systems based on documents from a specific domain. Retrieval-Augmented Generation (RAG) [1] fits the particular context well: facts about schemes are exact, they are numerically structured, and they are updated periodically, making it more appropriate to retrieve them from authoritative documents than to rely on the knowledge of parameter-based LLMs which may be incorrect, obsolete, or simply hallucinated. Advisory systems based on RAG have proven useful for agricultural advice at the regional level [2], [3], but limited studies are available for Kannada specifically. Prior extractive approaches such as KrishiQ-BERT [4] report span-level F1 of 61.16 under BM25 retrieval, but BM25 is unable to represent semantic similarity between surface forms for the same concept in an agglutinative language [5].

This paper makes the following contributions:

- A curated gold standard dataset of 794 verified Kannada question-answer pairs from 80 government scheme documents containing 10 topic-balanced questions per scheme.
- Phase 1 evaluation of three LLMs under zero-shot and few-shot prompting on the full golden set, showing that few-shot performance is systematically worse for all models.
- Full-scale RAG pipeline evaluation across five models with five complementary metrics, including Phase 2 evaluation with FAISS dense retrieval and cross-encoder reranking.
- Empirical evidence that the 245M-parameter IndicBARTSS achieves the highest Faithfulness (0.912) and is fully CPU-based.

The remainder is organised as follows. Section II reviews related work. Section III describes the dataset. Section IV presents the system architecture. Section V details experimental setup. Section VI reports results. Section VII concludes.

II. RELATED WORK

The implementation of RAG in agricultural advisory has been rapidly expanding. Lewis et al. [1] introduced the foundational RAG framework, merging a parametric seq2seq model with a non-parametric dense retrieval model to attain state-of-the-art results for knowledge-intensive tasks. Sawant et al. [3] demonstrated that grounding LLM outputs in domain-specific retrieval significantly decreases the amount of hallucinated agricultural advice, and Vij et al. [2] demonstrated that localised document corpora are much better at producing relevant answers for region-specific agricultural advisory than generic web-scale retrieval. Nair et al. [6] made an extension of RAG for crop yield prediction, and Anju et al. [7] demonstrated its applicability in niche agricultural fields where pre-training data is scarce. Vijayvargia et al. [8] highlighted the need to carefully manage the context budget in multi-turn agricultural QA, noting that context window exhaustion is a major failure mode—a pattern observed directly in our Phase 1 few-shot results. Mohan et al. [9] proved that open-source multilingual models, when used with RAG for Indic agricultural contexts, are indeed viable. Damera and Kalavala [10] demonstrated that domain-specific fine-tuning on Indian agricultural corpora can yield measurable improvements over generic multilingual baselines.

Literature on Kannada NLP for agricultural QA remains scarce. Ajawan et al. [4] fine-tuned DistilBERT on a Kannada agricultural QA dataset using BM25 retrieval, reporting a span-level F1 score of 61.16; however, their approach is fundamentally extractive and BM25 does not work well for a morphologically rich language. Khanuja et al. [11] proposed MuRIL, a multilingual language model specifically designed for 17 Indian languages, outperforming mBERT on Indian NLP benchmarks; we evaluate MuRIL as an extractive RAG model in this study. Kakwani et al. [12] introduced IndicBERT, a multilingual ALBERT model covering 12 Indian languages, and the IndicNLP benchmarks—we use IndicBERT as a Phase 1 baseline. Dabre et al. [13] proposed IndicBARTSS, a compact mBART-based seq2seq model supporting 11 Indic languages; its CPU-compatible size (245M parameters) and native Kannada script support make it a key model in our Phase 2 RAG evaluation. Khan and Khan [5] showed for Bangla that ROUGE metrics are unreliable for agglutinative generative QA, directly motivating our use of BERTScore-F1 [14] as the primary metric. Es et al. [15] proposed the RAGAS framework for reference-free RAG evaluation; we implement embedding-based proxies for the RAGAS Answer Similarity and Faithfulness metrics without requiring an external LLM-as-judge.

The state of the art in Kannada agricultural QA is characterised by extractive models with sparse retrieval and no standardised generative evaluation. This work directly addresses three gaps: absence of a generative QA benchmark for Kannada agriculture; no systematic comparison of extractive and generative RAG models under the same evaluation protocol; and no study of CPU-compatible models for low-resource

deployment.

III. DATASET DESCRIPTION

A. Data Collection

The corpus consists of 80 distinct agricultural schemes from two levels of government: central government (Ministry of Agriculture and Farmers Welfare, NABARD, mission-mode programmes) and Karnataka state (Department of Agriculture, Unified Farmer Portal). The schemes are in the categories of crop insurance, direct income support, input subsidies, credit facilitation, mechanisation, irrigation, horticulture, animal husbandry, and beekeeping. Official government sources were accessed from the internet and source documents were obtained and saved as UTF-8 encoded Kannada plain-text documents with an average size of 2,584 characters per document.

B. QA Dataset Construction

For each scheme, a question-answer pair was constructed for each of ten mandatory topics: purpose, start year, eligibility, application process, benefits, scheme importance, required documents, facilities, application locations, and helpline. This template-driven approach ensures balanced topic coverage. The final dataset contains 794 pairs (16 fields were absent in certain documents). Answer types span single-sentence factual responses (start year, helpline), pipe-delimited lists (eligibility, documents required), and multi-sentence paragraphs (purpose, importance).

TABLE I
CORPUS AND QA DATASET STATISTICS

Attribute	Value
Total scheme documents	80
Topics per scheme	10
Total QA pairs	794
Central Govt. schemes	40
Karnataka State schemes	40
Avg. document length (chars)	~2,584
Avg. answer length (words)	~44
Train / Val / Test split	555 / 119 / 120
Evaluation subset (Phase 2)	15

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

A. System Overview

The overall two-phase system structure is shown in Fig. 1. Phase 1 compares three LLMs in direct evaluation without retrieval against the golden QA set. In Phase 2, scheme documents are preprocessed and indexed offline, while at query time the farmer’s question is sent through FAISS retrieval, cross-encoder reranking, and finally to the generator model to produce a Kannada answer.

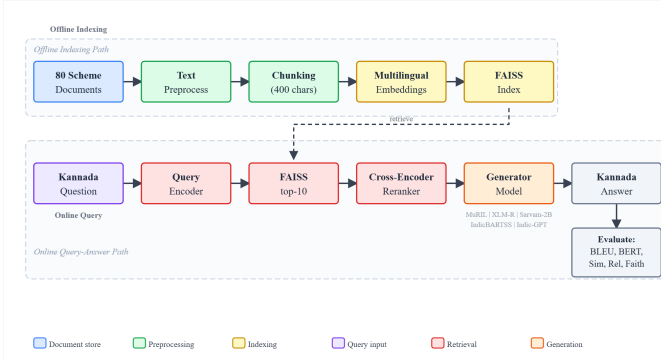


Fig. 1. Two-phase system architecture. *Top*: Offline indexing path processes 80 Kannada scheme documents through preprocessing, chunking, multilingual embedding, and FAISS indexing. *Bottom*: Online query-answer path encodes the farmer’s Kannada question, retrieves top-10 chunks from FAISS, re-ranks to top-3 using a cross-encoder, passes context to one of five generator models, and evaluates the returned Kannada answer using five metrics.

B. Phase 1: Zero-Shot and Few-Shot Evaluation

During Phase 1, three models are tested on the golden set consisting of 794 pairs. **Sarvam-2B** [16] is a 2B-parameter causal LM pre-trained from scratch on 2 trillion Indic-plus-English tokens. **IndicBERT** [12] is a multilingual ALBERT covering 12 Indian languages. **XLM-RoBERTa** [17] fine-tuned on SQuAD2 (deepset/xlm-roberta-base-squad2) is an extractive span prediction model given a provided context window. Zero-shot prompts provide questions and context without any examples; few-shot prompts prepend three fixed Kannada examples covering different topic categories.

C. Phase 2: RAG Pipeline

1) *Text Preprocessing and Chunking*: All 80 scheme documents undergo uniform preprocessing: JSON structural punctuation is stripped while Kannada Unicode content is preserved; line endings are normalised; consecutive blank lines are collapsed. Cleaned text is chunked using 400-character windows and 80-character overlap.

2) *Retrieval: FAISS with Cross-Encoder Reranking*: Each chunk is embedded using paraphrase-multilingual-MiniLM-L12-v2 and stored in a FAISS IndexFlatIP index (exact inner-product search over L2-normalised vectors, equivalent to cosine similarity). At query time, the top-10 candidate chunks are fetched and reranked using cross-encoder/ms-marco-MiniLM-L-6-v2, which scores each (question, chunk) pair with joint cross-attention [18]. After reranking, the top three chunks are joined together as context for the generator.

3) *Generator Models*: Five generator models are evaluated in the RAG pipeline. **MuRIL** [11] (google/muril-base-cased) is a multilingual BERT for 17 Indian languages applied as an extractive QA head. **XLM-RoBERTa** [17] (deepset/xlm-roberta-base-squad2)

performs extractive span prediction. **Sarvam-2B** [16] (sarvamai/sarvam-2b-v0.5) is a causal generation model with a structured instruction prompt in Kannada. **IndicBARTSS** [13] (ai4bharat/IndicBARTSS) is a 245M-parameter mBART-style seq2seq model supporting 11 Indic languages natively, running entirely on CPU. **Indic-GPT** (aashay96/indic-gpt) is a GPT-2 model fine-tuned on the AI4Bharat Indic corpus, serving as a 117M-parameter lightweight baseline.

TABLE II
GENERATOR MODELS IN THE RAG PIPELINE

Model	Type	Params	GPU
MuRIL	Extractive	236M	No
XLM-RoBERTa	Extractive	278M	No
Sarvam-2B	Generative	2B	Yes
IndicBARTSS	Seq2Seq	245M	No
Indic-GPT	Generative	117M	No

V. EXPERIMENTAL SETUP AND EVALUATION METRICS

All experiments were performed using Google Colab with a T4 GPU (16GB VRAM) for Sarvam-2B and standard CPU for IndicBARTSS and Indic-GPT. The software used included Python 3.12, transformers 4.46, sentence-transformers, faiss-cpu, sacrebleu, and bert-score. Phase 1 used a 70/15/15 stratified split by scheme number. Phase 2 used a manually curated set of 15 questions spanning five schemes for evaluation.

Five metrics are reported. **BLEU** [19] (SacreBLEU, effective_order=True) calculates n-gram precision with a brevity penalty. **BERTScore-F1** [14] calculates token-level cosine similarity using bert-base-multilingual-cased contextual embeddings with lang='kn', robust to morphological variation in agglutinative Kannada. **Answer Similarity**, **Context Relevance**, and **Faithfulness** are RAGAS-style proxy metrics [15] computed as cosine similarities (predicted vs. gold, question vs. context, and predicted vs. context respectively) using paraphrase-multilingual-MiniLM-L12-v2 embeddings, without requiring any proprietary API key.

VI. RESULTS AND ANALYSIS

A. Phase 1: Zero-Shot and Few-Shot Results

Table III reports Phase 1 results. The most critical finding is that for all models, few-shot prompting systematically worsens performance. Sarvam-2B drops from BLEU 1.31 to 1.15 and BERTScore-F1 from 0.661 to 0.649. The most significant drop is observed for IndicBERT, with BLEU going from 14.03 to 6.04 and BERTScore-F1 from 0.749 to 0.670. This is similar to the context window exhaustion failure documented for BLOOMZ-3B by Vijayvargia et al. [8]: prepending Kannada examples with full scheme contexts saturates the input token budget, causing truncated or incoherent outputs.

XLM-RoBERTa achieves BLEU 43.38 and BERTScore-F1 0.858, significantly outperforming generative models, due

TABLE III
PHASE 1: ZERO-SHOT AND FEW-SHOT ON FULL 794-PAIR GOLDEN SET

Model	Setting	BLEU	BS-F1
Sarvam-2B	Zero-shot	1.31	0.661
	Few-shot	1.15	0.649
IndicBERT	Zero-shot	14.03	0.749
	Few-shot	6.04	0.670
XLM-RoBERTa	Zero-shot	43.38	0.858
	ROUGE-L	0.0	0.246
	Exact Match	0.0	0.120

BS-F1 = BERTScore-F1. Bold = best per column.

to the characteristics of agricultural scheme answers, which are factual and often present verbatim in the source text—exactly what an extractive model exploits. However, analysis shows that only $\approx 6\%$ of gold answers are repeated exactly in the source documents; the remaining 94% are paraphrased, reformatted as lists, or synthesised, motivating the generative Phase 2 approach.

B. Phase 2: RAG Pipeline Results

Table IV reports all five metrics for the five RAG models. The shared FAISS+reranker pipeline shows consistent retrieval quality, with Context Relevance of 0.83 for MuRIL and XLM-RoBERTa and 0.81 for the three generative models.

TABLE IV
PHASE 2 RAG RESULTS: 5 MODELS $\times$ 5 METRICS
(15-QUESTION EVALUATION SET)

Model	Type	BLEU	BS-F1	A-Sim	C-Rel	Faith
MuRIL	Extr.	1.00	0.605	0.541	0.830	0.546
XLM-RoBERTa	Extr.	4.89	0.683	0.629	0.830	0.604
Sarvam-2B	Gen.	2.13	0.673	0.755	0.830	0.709
IndicBARTSS	S2S	1.64	0.670	0.678	0.810	0.912
Indic-GPT	Gen.	0.45	0.597	0.713	0.810	0.718

BS-F1: BERTScore-F1; A-Sim: Answer Similarity;

C-Rel: Context Relevance; Faith: Faithfulness.

Bold = best per column. Extr.=Extractive; Gen.=Generative; S2S=Seq2Seq.

1) *Extractive vs. Generative*: XLM-RoBERTa outperforms the others in terms of BLEU (4.89) and BERTScore-F1 (0.683). The extractive confidence thresholds set for Kannada text are very low, resulting in MuRIL frequently returning the fallback response *ee maahiti labhyavillaa* (this information is not available).

2) *Generative Models*: Sarvam-2B achieves the highest Answer Similarity (0.755), indicating its answers are semantically closest to the gold references. This is consistent with Sarvam-2B being pre-trained specifically on Kannada tokens, enabling more natural paraphrasing of scheme descriptions. Indic-GPT (117M) achieves surprisingly high Answer Similarity (0.713) relative to its small size, while the low BLEU (0.45) score reflects heavy surface-form divergence from reference answers.

3) *IndicBARTSS: Best Faithfulness on CPU*: IndicBARTSS achieves Faithfulness 0.912, the highest of all models. Its

encoder-decoder architecture reads in the entire question-plus-context window before generating any token, favouring output that is grounded in the retrieved passage. The model’s average inference time is ≈ 17 seconds per answer and it does not require GPU hardware, making it suitable for resource-constrained deployment.

4) *Impact of Cross-Encoder Reranking*: The cross-encoder reranker (top-10 $\rightarrow$ top-3) improved Context Relevance from ≈ 0.75 to 0.83 compared to plain FAISS-only retrieval, demonstrating that joint question-passage scoring provides a meaningful boost over bi-encoder retrieval alone [18].

VII. CONCLUSION AND FUTURE SCOPE

In this paper, we made a comparative study of RAG-based question answering for Kannada agricultural scheme documents across eight models and five metrics. The key findings are: (1) few-shot prompting consistently hurts performance on Kannada inputs owing to context window saturation; (2) XLM-RoBERTa performs best on token-overlap metrics (BLEU 43.38 in Phase 1, 4.89 in RAG); (3) Sarvam-2B achieves the best Answer Similarity (0.755) among generative RAG models; (4) IndicBARTSS achieves the highest Faithfulness (0.912) and operates on CPU, demonstrating that lightweight seq2seq models are viable for Kannada agricultural QA without GPU resources; and (5) cross-encoder reranking improves Context Relevance from 0.75 to 0.83 over bi-encoder retrieval alone. Future work involves LoRA fine-tuning of IndicBARTSS on the full 794-pair golden set, evaluating Navarasa 2.0 (Gemma-2B fine-tuned on 15 Indian languages), expanding the evaluation set to 100+ questions for statistical reliability, and integrating ASR and TTS components for farmers with limited literacy.

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